T-L-411C
October 9, 1970
SUPERSEDING
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May 1, 1967

FEDERAL SPECIFICATION

LINES: SHOT (FOR) LINE-THROWING-GUNS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers shot-line for line-throwing-guns.

1.2 Classification.

1.2.1 Types and classes. The shot-lines covered by this specification shall be of the following types, classes, and sizes (diameter) as specified (see table I and 6.2):

Type I - Flax, cable laid.

Class 1 - Natural finish.

Type II - Nylon, braided without core.

Class 2 - Waxed finish.

Type III - Nylon, cable laid.

Class 2 - Waxed finish.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

UU-T-81 - Tags, Shipping and Stock.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-C-3131 - Cordage; Preparation for Delivery of.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Type I line shall be fabricated from flax fiber. Types II and III lines shall be fabricated from multifilament nylon.

3.2 Standard sample. When a standard sample is established, the dyed line shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

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3.3 Color. Unless otherwise specified, the color of the line shall be natural. When a specific color is required, it shall be as specified (see 6.2) and shall match the standard sample for the applicable shade.

3.3.1 Colorfastness. The dyed line shall show colorfastness to weathering equal to or better than the standard sample (see 6.3) when tested as specified in 4.2.4. When no standard sample has been established, the dyed line shall show fair colorfastness to weathering when tested as specified in 4.2.4.

3.4 Construction and physical requirements. Types I and III lines shall be cable laid construction. The lay for types I and III lines shall be right hand or "Z" lay. Type II line shall be braided without core construction. The finished lines shall conform to the requirements specified in table I when tested as specified in 4.2.4. The loss in breaking strength after heat aging for types II and III lines shall not exceed 10 percent of the original breaking strength when tested as specified in 4.2.4.

TABLE I. Physical requirements

Type and class	Diameter (± 1/64 inch)	Number of strands	Length per pound (min. feet)	Breaking strength (min. pounds)	Elongation load pounds	Elongation (max. percent)
Type I, class 1	1/8	9 1/	175	350	260	15
	7/32	9 1/	75	700	520	15
	9/32	9 1/	45	1150	850	15
Type II, class 2	3/32	8	400	415	370	15
Type III, class 2	13/64	9 1/	100	940	900	30

1/ Cable laid construction with 3 opposite twist strands laid together to form 9 strands (3 rope structure).

3.5 Finish. The class 1 line shall have a soft natural finish. The class 2 line shall have a smooth, polished, waxed finish.

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3.6 Nonfibrous material. The nonfibrous material content of the type I, class 1 line shall not exceed 8.0 percent when tested as specified in 4.2.4. The chloroform soluble material of the finished types II and III lines shall not exceed 10.0 percent when tested as specified in 4.2.4.

3.7 Put-up. Unless otherwise specified (see 6.2), the lines shall be furnished in coils in the minimum lengths specified in table II. A plus tolerance of 2 percent will be allowed on the length of the coil specified. Each coil shall be continuous throughout and shall contain no splices, knots, or loose ends, and shall be so wound that each turn and layer is free from entanglement. The ends of all lines shall be securely served, or whipped, except that types II and III line ends may be heat sealed to prevent unlaying or fraying of line. All fag ends shall be cut off.

TABLE II. Length

Type and class	Diameter (inch)	Length of coil 1/ (min. feet)
Type I, class 1	1/8	2000
	7/32	2000
	9/32	2000
Type II, class 2	3/32	500
Type III, class 2	13/64	2000

1/ See 4.2.2.

3.8 Identification. Each coil of line shall have a ticket (identification tag) attached to it for identification purposes. The ticket shall conform to the requirements for type B, class 1, size 4 or 5 of UU-T-81. The ticket shall be made of not less than 15-point paper stock and shall have a minimum tearing resistance of both directions (total) of 850 grams when tested as specified in UU-T-81. The ticket shall be legibly printed, stamped or typed with water insoluble black ink. Handwritten entries shall be prohibited. The ticket shall contain the following information:

Stock number
Nomenclature
Specification number
Length
Type, class, and size (diameter)
Contract number and date
Supplier's name

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3.9 Workmanship. The finished line shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination of the end item for visual defects. Defects found during this examination shall be classified in accordance with the defects listed in table III, regardless of their proximity to each other. The sample unit for this examination shall be one coil. Ten percent of the length contained on each sample unit, but not less than 100 feet, shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of coils each. The inspection level shall be level III. The acceptable quality level (AQL) shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. If one or more critical defects are found in the sample, the lot shall be unacceptable.

TABLE III. Visual examination

Examine	Defect	Classification		
		Critical	Major	Minor
Appearance and workmanship	Any cut.	X		
	Chafed or damaged.	X		
	Kink, knot, broken or loose end or strand.	X		
	Knotted, spliced, or otherwise joined to make a continuous length.	X		

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TABLE III. Visual examination (cont'd)

Examine	Defect	Classification		
		Critical	Major	Minor
Appearance and workmanship (cont'd)	Line not of one continuous length.	X		
	Improperly or not firmly wound resulting in kinking, knotting, entangling, or slippage during unwinding or otherwise affecting free, unhampered unwinding of line.	X		
	Put up (coils) not as specified.	X		
	Fag ends not cut off.		X	
	Any end not heat sealed, whipped or served.		X	
Finish:				
(class 1)	Not soft natural finish.			X
(class 2)	Not smooth and polished.			X
	Wax coating shows lumps.			X
Type II	Unevenly braided resulting in open places, breaks in continuity of braid, or soft spots.	X		
Color (when applicable)	Other than specified.		X	
Type	Other than specified.		X	
Cleanness	Spot or stain clearly visible. 1/			X
Identification	Omitted, incorrect, illegible, insecurely attached.			X
	Ticket not as specified.			X
	Handwritten entries.			X

1/ At normal inspection distance (approximately 3 feet).

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4.2.2 Examination for length. The sample unit for this examination shall be one coil. The inspection level shall be level S-3 and the acceptable quality level shall be 4.0 percent defective. For lots consisting of 500 or fewer coils, the sample size shall be 10 and the acceptable number 1. The lot size shall be the number of coils in the inspection lot. Defects shall be as listed in 4.2.2.1. The length on the coil (see table II) shall be determined using the following formula:

W x L = Length per coil in feet.

Where: "W" is the net weight of the coil in pounds (to the nearest 0.1 pounds).

"L" is the lot average of the weight -- length relationship (length/pound) as determined in 4.2.4.

4.2.2.1 Defects with regard to length. Defects shall be considered to exist if any of the following are determined during inspection:

(a) The calculated length of coil is less than the minimum specified or more than 2 percent in excess of the length specified in table II.

(b) The length of coil marked on the ticket is more than 6 feet less than specified in table II.

4.2.3 Examination of preparation for delivery requirements. An examination shall be made to determine whether packaging and packing complies with the section 5 requirements of this specification. The examination shall be in accordance with the provisions of MIL-C-3131, except that the inspection level shall be S-2 and AQL shall be 2.5 defects per 100 units.

4.2.4 Testing of the end item. The methods of testing specified in Fed Test Method Std. No. 191 wherever applicable and as listed in table V shall be followed. The physical and chemical values specified in section 3 except where otherwise specified, apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample size shall be in accordance with table IV. The sample unit for testing shall be one coil. The lot size shall be expressed in units of coils. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions, except in settlement of dispute, in which case the tests shall be made upon material which has reached equilibrium under standard conditions as defined in Fed. Test Method Std. No. 191.

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TABLE IV. Sampling for tests

Lot size (coils)	Sample size
15 and under	2
16 up to and including 50	3
51 up to and including 150	5
151 up to and including 280	8
281 up to and including 500	13
501 and over	20

TABLE V. Test method

Characteristic	Requirement reference	Test method	No. of deter. per sample unit	Results reported as
Material:				
Type I	3.1	1210 1/	-	-
Types II and III	3.1	1530 1/	-	-
Color matching	3.3	4.2.4.2	1	Pass or fail
Colorfastness to weathering	3.3.1	5671 2/	1	Pass or fail
Number of strands	3.4	Visual	1	Pass or fail
Diameter	3.4	6002	5	Avg. of 5 deter. to nearest 1/64 inch
Direction of twist (types I and III)	3.4	4050	-	Pass or fail
Length per pound (types I and III)	3.4	6004 4/	3	Avg. of 3 deter. to nearest 0.1 foot per pound

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TABLE V. Test method (cont'd)

Characteristic	Requirement reference	Test method	No. of deter. per sample unit	Results reported as
Length per pound (type II)	3.4	4010 4/	3	Avg. of 3 deter. to nearest 0.1 foot per pound.
Breaking strength:				
Initial	3.4	4102	5	Avg. of 5 deter. to nearest 0.5 pound
After aging (types II and III)	3.4	4102 and 4.2.4.1	5	Avg. of 5 deter. to nearest 0.5 pound
Elongation	3.4	4102 3/	5	Avg. of 5 deter. to nearest 0.2 percent
Nonfibrous material (type I)	3.6	2620 1/	-	-
Chloroform soluble material (types II and III)	3.6	2611 1/	-	-

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The specimens shall be exposed for 40 standard fading hours.

3/ Elongation shall be determined at the specified load in table I.

4/ Except that the result shall be converted to feet per pound.

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4.2.4.1 Determination of breaking strength after heat aging (type II and III). Five breaking strength specimens prepared as specified in method 4102 shall be heated for 5 days in a convection air oven at 175° ± 2°F. After aging, the specimens shall be allowed to reach equilibrium under Standard Conditions before being tested for strength. The percent change in breaking strength shall be calculated based on the average breaking strength originally and after heat aging.

4.2.4.2 Color matching. The shade of the dyed finished line shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin (see 6.3).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The shot-line, put-up as specified, shall be packaged in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The shot-line shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, shipments shall be marked in accordance with MIL-C-3131.

6. NOTES

6.1 Intended use. Shot-line covered by the specification is intended for use in the operation of line throwing guns.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, class, and size (diameter) required (see 1.2.1 and 3.4).
(c) When color required is other than natural (see 3.3).
(d) Put-up if other than specified (see 3.7).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to the standard sample, address the procuring office issuing the invitation for bids.

MILITARY CUSTODIANS:

Army - GL
Navy - SH
Air Force - 84

Review activities:

Army - MEC
Navy - AS, OS

User activities:

Navy - CG
Air Force - 26, 82

Preparing activity:

Army - GL

CIVIL AGENCY INTEREST:

COM-NBS
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